ETC3250/5250 Introduction to Machine Learning

Week 11: Evaluating your clustering model

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Overview

We will cover:

- Confusion tables
- Cluster metrics
- Numerical summaries of solution
- Visual summaries
- Low-dimensional representations

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Evaluating your clustering

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Confusion tables: comparing results (1/4)

The dendrograms look similar. Are the clusters similar?

- Both have reasonable support for 2 clusters
- M1 has support for 3 clusters while M2 supports 4
- M2 has support for 6 clusters and M1 support 9
- Is the 2-cluster solution from M1 and M2 the same?
- Two clusterings can be compared with a confusion matrix, similarly to supervised classification.
- However, here we are comparing methods to each opther rather than to the truth
- The main difference though is cluster labels are not consistent between methods. Comparison is better if the labels are matched optimally, as a first step.

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Confusion tables: comparing results (2/4)

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Confusion tables: comparing results (3/4)

We compare the 2-cluster solutions

```
# A tibble: 2 × 3
  cl1     `1`   `2`
  <fct> <int> <int>
1 1        30    36
2 2        20    14
```

Match labels: What method 1 calls cluster 1, method 2 calls cluster 2. Rearrange the confusion matrix, or re-label one methods clusters, so that large counts are on (top-left to bottom-right) diagonal.

```
# A tibble: 2 × 3
# Groups:   cl1 [2]
  cl1     `1`   `2`
  <fct> <int> <int>
1 2        20    14
2 1        30    36
```

Do we now think these clusters are similar or different?

The two clusterings are indeed quite different. If you look at the model coefficients, it is clear.

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Confusion tables: comparing results (4/4)

For the penguins data

- Average Linkage: 2 or 5 clusters?
- Wards Linkage: 2, 3 or 5 clusters?

Comparing the 5 cluster solutions (note reordering)

```
# A tibble: 5 × 6
  cl1     `1`   `2`   `5`   `3`   `4`
  <fct> <int> <int> <int> <int> <int>
1 1       102    52     0     0     0
2 2         3     5    56     0     0
3 3         0     0     0    52    67
4 4         0     0     0     0     4
5 5         0     0     1     0     0
```

- You might think that average linkage would give similar results to Wards linkage, but results here are very different.

- Average divides the data into three large clusters with two clusters with few observations.
- Wards divides the data into five large clusters, with one having twice the points as the others.

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Cluster metrics (1/2)

We covered: within.cluster.ss, WBRatio, Hubert Gamma, Dunn, Calinski-Harabasz Index.
They measure:

In an idealistic setting, these are all perfect metrics.
Remember the world has many shapes:

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Cluster metrics (2/2)

Clustering algorithms not primarily looking for gaps. Gaps can be local or global in size, or just not exist. Clustering that partitions data can still be useful. Know how an algorithm will partition.

What would happen to these data sets?

The shape (strictly correlation here) strongly affects how an algorithm divides or partitions a blob.

k-means is always drawing circles

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Numerical summaries

- Means: which variables differentiate that variable from another
- Standard deviations: how tightly packed is that cluster
- Variances and covariances: but unwieldy to print all
- Cluster size

Wards linkage hierarchical clustering

		Clusters		
Var	**Stat**	**1**	**2**	**3**
id	mean	-0.87	0.66	1.07

Clusters

Var	Stat	1	2	3
	sd	*0.54*	*0.56*	*0.50*
bd	mean	0.58	-1.10	0.73
	sd	*0.62*	*0.50*	*0.54*
fl	mean	-0.78	1.16	-0.27
	sd	*0.46*	*0.46*	*0.46*
bm	mean	-0.66	1.09	-0.49
	sd	*0.58*	*0.63*	*0.42*
	n	162.00	123.00	57.00

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Visual summaries

- Statistics: Show the mean and variances for each cluster, as a parallel coordinate plot.
- Model in the data space:
  - Colour observations by their cluster label, and use multivariate plots: Scatterplot matrix, parallel coordinates and tours
  - Dendrograms: Draw the connections of points being joined to form clusters as a high-d graph.
  - Ellipses: As done in summarising model-based fit.
  - Net: As done for SOM.
- Convex hulls: Bounding shapes of each cluster, can be done in high-D, too.

Statistics

Shading represents 1 standard deviation above and below the mean

- Cluster 1 has high values of `bl` and low values of `bd`, `fl` and `bm`.
- Cluster 2 has low values of `bl`, and high values on `bd`, `fl` and `bm`.
- Cluster 3 has high values of `bl`, `bd` and low values of `fl` and `bm`.

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Model-in-the-data-space (1/4)

The clusters are neatly distinguished in different pairs of variables.

Cluster 2 is neatly separated. Clusters 1 and 3 are distinct but there is no gap.

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Model-in-the-data-space (2/4)

The dendrogram can be drawn in the data space.

- Add extra points at average between clusters indicating joins.
- Create variable specifying point type, “data” or “node”.
- Add edges data, specifying which points are connected to make dendrogram.

Model-in-the-data-space (3/4)

Why?

Because it shows us how the algorithm is sequentially joining the observations, relative to their values.

Model-in-the-data-space (4/4)

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Convex hulls

A set $S \subset \mathbb{R}^d$ is convex if for any $x \in S$, $y \in S$ and $\alpha \in [0, 1]$ then $\alpha x + (1 - \alpha)y \in S$

Convex hulls are a common method for visually displaying the region in the data space corresponding to each cluster.

It avoids making any assumption about the shapes.

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Convex hulls

Convex hulls are defined in p dimensions also.

Are clusters 1 and 3 separated/distinct?

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Low-dimensional representations

- PCA, but not PCA
- Perhaps, projection pursuit (guided tour)
- Nonlinear dimension reduction
- Self-organising map

Lets look at the `c1` data from the `mulgar` package

```
Importance of components:
                          PC1   PC2    PC3    PC4     PC5     PC6
Standard deviation     2.802 1.720 0.4204 0.3470 0.05296 0.04880
Proportion of Variance 0.707 0.266 0.0159 0.0108 0.00025 0.00021
Cumulative Proportion  0.707 0.973 0.9887 0.9995 0.99979 1.00000
```

Proportion of total variance is 0.97 with 2 PCs. But if you only used 2 PCs, you would miss the clusters visible only with PC4.

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How do you know you got it right?

All “models” are wrong by some are useful.
Your clustering model is ‘right’, if it was useful

- Are the subsets useful, e.g. if you market differently to the different subsets do you get more purchases than marketing the same way to all.
- Can you reduce the number of parameters needed in a high-dimensional regression model, by grouping genomes into a smaller set?
- Does it match the shape of the data?
- Bootstrap or CV
- Domain knowledge and interpretability

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Wrapping up

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What we have covered

- Computational and statistical models for a categorical response: logistic regression, LDA, decision trees, forests, support vector machines, kNN, neural networks
- Discovering class labels by clustering: k-means, hierarchical, model-based, self-organising maps
- Dimension reduction: princi**pal** components, multidimensional scaling t-SNE, UMAP
- Visualising your high-dimensional data and models

Not covered

- *Continuous, numerical response*: $\longrightarrow$ ETC2420/5242, ETC3580/5580
  - Fit statistics easier: MSE, AIC, R^2, but not all applicable for categorical response
  - Visualisation a little harder
- *Algorithms for model fitting*: Logistic regression, neural networks and $\longrightarrow$ ETC3555/5555

- *Full range of neural networks*: how to prevent overfitting, convolutional, recurrent, word embeddings $\longrightarrow$ ETC3555/5555

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Philosophy

Some technicians brag that they can pull an appliance into parts. That is not as impressive as the technician who can put it back together, and make it work. I would hire this person in a heartbeat. *~Raymond Bruce Cook*

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The exam

End of semester exam counts for 60% of the module grade

You will have 2hrs and 10 mins to answer 6 Questions
This is a closed-book eExam with Specifically permitted items

- Physical Calculator
- Double Sided sheet of A4 (must be physical)

The exam will be in supervised exam conditions

Make sure you read the question carefully!

Exam

3 questions, each made up of approx 4 short essay questions
e.g.

- Here is the output of this model in R, what is it telling us?
- What do we learn about the structure in our data from this image?
- Why is this bit of methodology important for this model?
- What do you need to do to this data to fit this model?
- What's the difference between this model and this model?
- What complications are there fitting this model in stead of this model?
- What are the benefits of using this methodology over that methodology?
- Based on this output which model would you choose?

Short typed answers in eAssessment platform
3 hybrid questions, longer and made up of stages
e.g.

- Implement one iteration of this algorithm on this toy data

- Define this model
- Sketch this model
- Explain how this algorithm works and why its preferable to another algorithm
- Explain the process with which you would learn these parameters or hyperparameters

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Preparing for the exam

The exam is like the quizzes, tutorial and assignment exercises all together, without the programming elements.

Continue to use the forum

Exam consultation Thursday 12th June 9:30am-11:30am

Ways to prepare:

- *Review your quizzes*.
- *Review the tutorial and assignment materials*: what results mean, what problems are detected in the model, conceptual thinking, and explanations of results and diagnostics.
- **Resources** provided have several books with exercises at the end of chapters. Work your way through some where you feel you don't know the topic well enough.
- *Form a small study group*: make up questions for each other to tackle, compare and contrast your answers.

- *Try asking GAI questions*, and evaluate the answers as useful or misguided based on your learning.

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Next: What the ETC5250 students have learned from the kaggle challenge

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